User Manual

Team SolarEase (Microsoft: AI Powered Solar Microgrid)

Version number:

Team members: Hannah Xiao, Oulu Zhang, Samar Khan

Instructor name: Luyao Niu, Mark Licata

Industrial sponsor: Peeyush Kumar

March 12, 2026

Product Overview

SolarEase is an AI-powered software platform designed to help communities make more confident decisions about participating in solar projects. Today, many community organizations and local stakeholders struggle to evaluate solar opportunities because key information is fragmented and difficult to interpret. Financial returns depend on complex variables such as energy pricing, incentives, and system performance, while power purchase agreements (PPAs) often contain dense legal and financial terms that are hard for non-experts to understand.

SolarEase addresses this challenge by bringing key information into one place and translating complex data into clear, actionable insights. The platform is designed for community leaders who facilitate group decisions and transparent negotiations, as well as households who are deciding whether to participate in a shared solar agreement.

The platform includes three core features: an AI assistant that answers questions and explains solar concepts, an ROI analysis tool that estimates potential savings based on local data, and a contract summary generator that highlights key terms and risks in PPAs. Together, these tools help users evaluate options with greater transparency and confidence.

System Requirements

Software projects must specify supported operating systems, hardware requirements, dependencies, internet requirements, and any required user accounts. Hardware projects must describe power requirements, environmental conditions, compatibility constraints, and required tools. Hybrid systems must address both. If your project includes physical components, include a "Package Contents" section listing all included items.

Supported Operating Systems

SolarEase is a web-based software platform and can be installed on the following operating systems:

- macOS (recommended)
- Windows 10 or later
- Linux (Ubuntu 20.04 or later)

The application runs locally through a web browser and is compatible with modern browsers such as:

- Google Chrome
- Microsoft Edge
- Firefox
- Safari

Hardware Requirements

SolarEase does not require specialized hardware.

Software Dependencies

The following software must be installed before running SolarEase:

- Node.js 16+ with npm
- Python 3.9+ with pip

The system also relies on the following key Python libraries:

- cvxpy (≥ 1.3.0) – used for convex optimization in solar allocation calculations
- numpy (≥ 1.24.0) – used for numerical computations

All additional dependencies are installed automatically through:

npm install

pip install -r requirements.txt

Internet Requirements

An internet connection is required for:

- Accessing external data sources used in solar analysis
- Using AI-powered features (if enabled)
- Retrieving solar resource and energy data

The application itself runs locally once installed.

Required User Accounts

Some features of SolarEase require API access.

Optional accounts include:

- OpenAI API account - required to enable AI assistant features such as contract summarization and question answering

If API credentials are not provided, the platform can still run with limited functionality.

Detailed installation and setup instructions

Website is fully deployed at gix-solarease.com

Prerequisites:

Node.js v18 or higher

Python 3.9+

Git

PostgreSQL 14+

Step 1: Clone the Repository

>> git clone https://github.com/GIX-Luyao/final-codebase-solarease.git

>> cd final-codebase-solarease

Step 2: Install Node.js Dependencies

>> npm install

Step 3: Install Python Dependencies

pip install cvxpy numpy

Step 4: Configure Environment Variables

>> cp .env.example .env

Open .env in a text editor and configure:

OPENAI_API_KEY

AZURE_OPENAI_ENDPOINT

AZURE_OPENAI_KEY

AZURE_OPENAI_DEPLOYMENT

JWT_SECRET

DB_HOST

DB_PORT

DB_NAME

DB_USER

DB_PASSWORD

See .env.example for example setup file

Step 5: Set Up the Database

Create a PostgreSQL database named solarease

>> psql -U postgres -d solarease -f server/db/schema.sql

Step 6: Start the Application

>> npm run api

>> npm run dev

Step 7: Open the App

http://localhost:5173 on browser

Operating instructions section

Once system is set up on localhost users can use the following features:

1. Soli (AI Agent)

   Click the chat icon in the bottom-right corner. Ask questions about community solar.

2. ROI Calculator

   Navigate to "ROI Calculator" from the header. Follow the multi-step wizard.

3. Contract Transparency

   Navigate to "Contracts" from the header. Upload a solar contract in PDF or TXT format.

4. User Accounts

   Click "Login" or "Register" in the header. Registered users can save contract analyses and ROI calculations.

5. Shutting Down

   Press Ctrl+C in each terminal window to stop the frontend and backend servers.

Technical Specifications

Architecture:

- Frontend: React 18 with Vite build tool
- Backend: Node.js with Express.js REST API server
- AI Integration: OpenAI GPT-4o-min
- Nash Bargaining Solver: Python (CVXPY + NumPy)
- Database: PostgreSQL
- Authentication: JWT access tokens
- Deployment: Azure

Safety, Privacy, and Security

- All AI API calls are proxied through the backend server. Keys are stored in environment variables and never sent to the browser.
- User passwords are hashed with bcrypt before storage.
- JWT access tokens expire after a short period.
- Contract uploads are restricted to PDF and TXT files, with a 10MB size limit.
- Conversation data is stored in the browser's localStorage only, not external service used.

A troubleshooting section

- IF npm run api fails with "Cannot find module" THEN Run npm install again
- IF AI chatbot says "I'm having trouble connecting THEN Check that your .env file has a valid OPENAI_API_KEY
- IF Nash Bargaining returns an error THEN Verify Python 3.9+ is installed
- IF Port 3000 or 5173 already in use THEN Stop other processes on those ports, or change PORT in the .env file

Limitations

- No real time data as all calculations are done with user inputs and forecasting models
- All AI interactions and tools are in English only
- Clearing browser data resets all personalization memory
- Unusual contracts may not work in contract transparency, it's optimized for standard community solar agreements.
- Website may look limited on mobile version